Project 1&2 - Breast Cancer Analysis and Visualization

Use this document to record the write up for this project. Each partner must complete their own writeup in their own words, though your external sources can be the same for both partners.

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https://colab.research.google.com/drive/1g_TulG7yIi3pK5hVEG4T4SuD8Fh4BiKb#scrollTo=8VYkSTWNeGEB

Task 1: Clean up the Data

Describe the code you used to analyze the data. What erroneous data elements did you find? How did you clean up the data? How many patients are in your final data after it is all cleaned up? (paragraph)

We analyzed and cleaned the University of Wisconsin Hospitals Breast Cancer dataset using the Practice 2 notebook. First, I used `cancer.info()` to get an overview of the dataset, which showed 573 records and 31 columns. From this, I noticed that some columns, such as `radius_mean` and `texture_mean`, contained missing values. To handle this, I used `cancer_clean = cancer.dropna()` to remove any rows with empty or null values. After this step, the dataset had 571 records.

Next, I checked the `diagnosis` column using `print(cancer_clean['diagnosis'].value_counts())` and found 357 benign cases and 214 malignant cases. I also verified the dataset using `print("After removing bad diagnosis rows:", cancer_clean.shape)` to make sure there were no invalid labels, and the count remained at 571. After that, I kept only rows with valid numeric values using `cancer_clean = cancer_clean[(cancer_clean.select_dtypes(include='float64') >= 0).all(axis=1)]`, since medical measurements should not be negative. I confirmed this step with `print("After removing negative value rows:", cancer_clean.shape)`, which removed one additional row and left 570 records.

Overall, the dataset was reduced from 573 to 570 patient records after cleaning, making it ready for further analysis.

Task 2: Visualize the Data

Sample Description of Visualization 0

This is a sample visualization and does not count as one of the five you must create.

What the visualization shows: The above graph shows a scatter plot of two measurements of tumors. The measurement along the X-axis is called "radius_mean" in the database and is the average distance from the center of the tumor to the edge. Along the Y-axis the measurement is the "texture_mean" and this is the standard deviation of the grayness of the color of the tumor image. Each cancer tumor is given a different color if it is malignant or benign.

Why this might be useful: This visualization is useful in determining if together the radius_mean and texture_mean provide enough data to determine if a tumor is malignant or not. While these two measurements do separate the tumors well, more of the malignant tumors are in the upper right of the graph, there is still significant overlap. This means it would be difficult to diagnose tumors from these two measurements alone.

Something you would like to improve with the visualization: One thing that could make this visualization better would be to have each dot transparent. Currently some dots hide other dots, especially in the dense areas of the graph.

Any external sources for the code you used--simple URLs are fine: This visualization is a modified version from the CIS 3115 Learning Practice 2 notebook.

Description of Visualization 3

What the visualization shows:

This visualization shows a scatter plot that is showing how the average tumor radius and texture relate in the breast cancer dataset. Each dot stands for a patient record, The x-axis shows the tumor's average radius, and the y-axis shows its average texture. I used color to highlight the diagnosis, making it easier to compare malignant and benign tumors.

Why this might be useful:

This chart helps spot patterns between tumor size and texture. It lets us see if larger tumors have different texture values and if malignant tumors group differently from benign ones. This can help researchers and healthcare workers understand which features are associated with each diagnosis.

Something you would like to improve with the visualization:

Something I would improve on this chart is to add trend lines or include additional tumor features, such as perimeter or area. This would help show how the variables relate. Making the chart interactive would also let users hover over each point to see more details about each patient.

Any external sources for the code you used:

- https://seaborn.pydata.org/generated/seaborn.scatterplot.html
- https://matplotlib.org/stable/index.html
- https://pandas.pydata.org/docs/
- https://colab.research.google.com/drive/1LP0ptbVn-d_GFj-bIfFssmTh5LZ0zlfR?usp=sharing

Description of Visualization 4

What the visualization shows:

This visualization shows a count plot of the number of cases for each diagnosis in the breast cancer dataset. The x-axis lists the diagnosis types, Benign and Malignant, and the y-axis shows the number of records in each group. This chart gives a quick overview of how diagnoses are distributed in the dataset.

Why this might be useful:

This chart is helpful because it shows whether the dataset is balanced or if one diagnosis happens more often than the other. Knowing how diagnoses are distributed is important before analyzing the data or building models, since an uneven split can affect the results.

Something you would like to improve with the visualization:

To improve this chart, I could add labels above each bar to show the exact counts. Using different colors for each diagnosis and adding a title and axis labels would also make the chart clearer and more attractive.

Any external sources for the code you used:

- https://seaborn.pydata.org/generated/seaborn.countplot.html
- https://matplotlib.org/stable/tutorials/index.html
- https://pandas.pydata.org/docs/
- https://colab.research.google.com/drive/1LP0ptbVn-d_GFj-bIfFssmTh5LZ0zlfR?usp=sharing

Description of Visualization 5

What the visualization shows:

This bar chart shows how many breast cancer cases fall into each diagnosis category in the dataset. I used the value_counts() function to count the number of benign (B) and malignant (M) tumors, then showed the results in the chart. The x-axis lists the diagnosis categories, and the y-axis shows the total number of cases for each.

Why this might be useful:

This chart is useful because it quickly shows how diagnoses are distributed across the dataset. I can easily see which diagnosis is most common. Knowing whether the dataset is balanced is important before conducting further analysis, as it can affect how results are interpreted and how models perform.

Something you would like to improve with the visualization:

To improve this chart, I could use different colors for each diagnosis to make it look better. Adding the exact counts above each bar would make the numbers easier to read. I could also make the labels and title bigger so the chart is easier to understand.

Any external sources for the code you used:

- https://pandas.pydata.org/docs/reference/api/pandas.Series.plot.html
- https://matplotlib.org/stable/tutorials/index.html
- https://pandas.pydata.org/docs/reference/api/pandas.Series.value_counts.html
- https://colab.research.google.com/drive/1LP0ptbVn-d_GFj-bIfFssmTh5LZ0zIfR?usp=sharing

My observation:

When I made this chart, I saw that there were more benign tumor cases than malignant ones in the dataset. This helped me understand the data better before looking at other variables.